4.4 Taylor's Theorem & The Convergence of Taylor Series

So far we have Taylor polynomials to approximate curves. In the last section we saw that these Taylor polynomials had intervals on which they "converged" or aligned with the desired function very well. The obvious question to ask from here is: When does a Taylor series of a function converge to the *entire* function?

Definition: Suppose f is differentiable n times at x_0. The nth-degree Taylor polynomial of f at x_0 is $p_n(x)=\sum_{k=0}^{n}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$ and $R_n(x)=f(x)-p_n(x)$ is called the n^{th} remainder of the Taylor series.

Theorem:

Proof:

From the above theorem we see that the key to showing that a function is equal to its Taylor series, is showing that $\lim_{n\to\infty} R_n(x)=0$ for appropriate values of x. So, we need a theorem that provides information about $R_n(x)$. We will call this theorem Taylor's Theorem.

Theorem (Taylor's Theorem also referred to as Taylor's Inequality):

Note: This theorem goes by many different names: Taylor's Theorem, Taylor's inequality, & The Remainder Estimation Theorem.

Note: In this theorem n represents the degree of the Taylor polynomial and not merely the final index value. This is important when working with Taylor series that have indexes that get rewritten such as with $\cos(x)$, $\sin(x)$, $\cosh(x), \sinh(x), etc.$

Example 1: Consider the function $f(x) = e^x$.

a) Use an appropriate Taylor polynomial of degree 2 to find a good estimate for e^1.

b) Use Taylor's Theorem to find an upper bound on the error of this approximation.

Example 2: Use Taylor's Inequality to determine a value of *n* you could use to approximate e accurately to two decimal places.

Example 3: Prove that $f(x) = e^x$ is equal to its Maclaurin series for all x.

Example 4: Assuming $x = 1$, what does example 1 tell us must be true?

Note: Using a partial sum of this series will only allow us to estimate the value of e and we can approximate the error of each approximation using Taylor's Theorem. However, the infinite series produces the exact value of e!

Example 5: Consider the function $f(x) = \sin(x)$.

a) Use the Maclaurin series for $f(x) = \sin(x)$, $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, to create a 7th degree polynomial to estimate the value of $\sin(1)$.

b) Use Taylor's Theorem to find an upper bound on the error of this approximation.

Example 6: Use Taylor's Inequality to determine a value of n you could use to approximate $\sin(1)$ accurately to 10 decimal places.

<u>Example 7:</u> Prove that $f(x) = \sin(x)$ is equal to its Maclaurin series for all x.